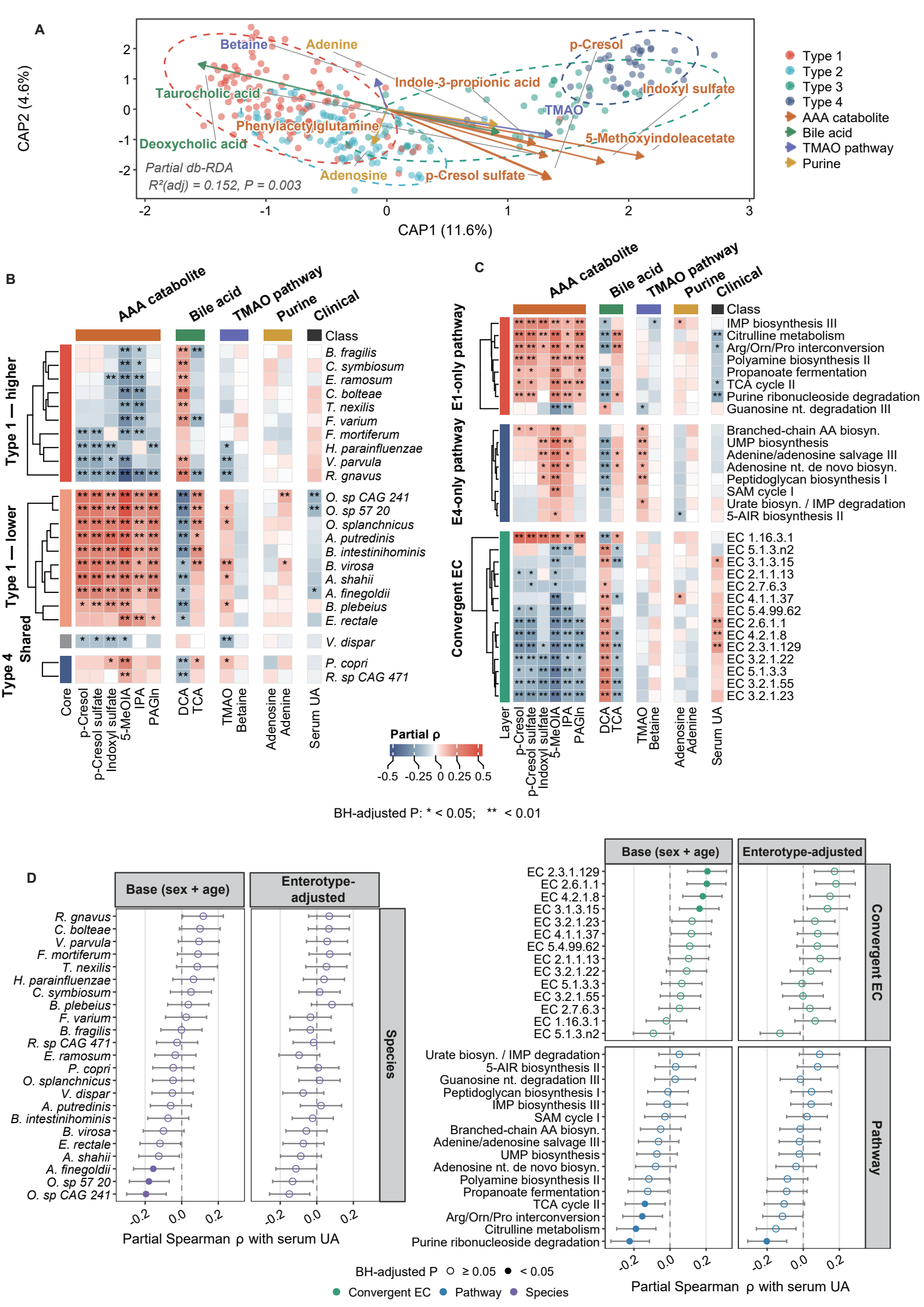

A
Betaine
Adenine
p-Cresol
Indole-3-propionic acid
Indoxyl sulfate
Taurocholic acid
TMAO
Phenylacetylglutamine
5-Methoxyindoleacetate
Deoxycholic acid
Partial db-RDA
R²(adj) = 0.152, P = 0.003
Adenosine
p-Cresol sulfate
CAP1 (11.6%)
CAP2 (4.6%)
Type 1
Type 2
Type 3
Type 4
AAA catabolite
Bile acid
TMAO pathway
Purine
B
AAA catabolite
Bile acid
TMAO pathway
Purine
Clinical
Class
Type 1 — higher
Type 1 — lower
Shared
Type 4
B. fragilis
C. symbiosum
E. ramosum
C. bolteae
T. nexilis
F. varium
F. mortiferum
H. parainfluenzae
V. parvula
R. gnavus
O. sp CAG 241
O. sp 57 20
O. splanchnicus
A. putredinis
B. intestinihominis
B. virosa
A. shahii
A. finegoldii
B. plebeius
E. rectale
V. dispar
P. copri
R. sp CAG 471
Core
p-Cresol
p-Cresol sulfate
Indoxyl sulfate
5-MeOIA
IPA
PAGln
DCA
TCA
TMAO
Betaine
Adenosine
Adenine
Serum UA
C
AAA catabolite
Bile acid
TMAO pathway
Purine
Clinical
Class
E1-only pathway
E4-only pathway
Convergent EC
IMP biosynthesis III
Citrulline metabolism
Arg/Orn/Pro interconversion
Polyamine biosynthesis II
Propanoate fermentation
TCA cycle II
Purine ribonucleoside degradation
Guanosine nt. degradation III
Branched-chain AA biosyn.
UMP biosynthesis
Adenine/adenosine salvage III
Adenosine nt. de novo biosyn.
Peptidoglycan biosynthesis I
SAM cycle I
Urate biosyn. / IMP degradation
5-AIR biosynthesis II
EC 1.16.3.1
EC 5.1.3.n2
EC 3.1.3.15
EC 2.1.1.13
EC 2.7.6.3
EC 4.1.1.37
EC 5.4.99.62
EC 2.6.1.1
EC 4.2.1.8
EC 2.3.1.129
EC 3.2.1.22
EC 5.1.3.3
EC 3.2.1.55
EC 3.2.1.23
Layer
p-Cresol
p-Cresol sulfate
Indoxyl sulfate
5-MeOIA
IPA
PAGln
DCA
TCA
TMAO
Betaine
Adenosine
Adenine
Serum UA
Partial ρ
-0.5 -0.25 0 0.25 0.5
BH-adjusted P: * < 0.05; ** < 0.01
D
Base (sex + age)
Enterotype-adjusted
Species
R. gnavus
C. bolteae
V. parvula
F. mortiferum
T. nexilis
H. parainfluenzae
C. symbiosum
B. plebeius
F. varium
B. fragilis
R. sp CAG 471
E. ramosum
P. copri
O. splanchnicus
V. dispar
A. putredinis
B. intestinihominis
B. virosa
E. rectale
A. shahii
A. finegoldii
O. sp 57 20
O. sp CAG 241
-0.2 0.0 0.2
Partial Spearman ρ with serum UA
Base (sex + age)
Enterotype-adjusted
Convergent EC
EC 2.3.1.129
EC 2.6.1.1
EC 4.2.1.8
EC 3.1.3.15
EC 3.2.1.23
EC 4.1.1.37
EC 5.4.99.62
EC 2.1.1.13
EC 3.2.1.22
EC 5.1.3.3
EC 3.2.1.55
EC 2.7.6.3
EC 1.16.3.1
EC 5.1.3.n2
Pathway
Urate biosyn. / IMP degradation
5-AIR biosynthesis II
Guanosine nt. degradation III
Peptidoglycan biosynthesis I
IMP biosynthesis III
SAM cycle I
Branched-chain AA biosyn.
Adenine/adenosine salvage III
UMP biosynthesis
Adenosine nt. de novo biosyn.
Polyamine biosynthesis II
Propanoate fermentation
TCA cycle II
Arg/Orn/Pro interconversion
Citrulline metabolism
Purine ribonucleoside degradation
-0.2 0.0 0.2
Partial Spearman ρ with serum UA
BH-adjusted P ○ ≥ 0.05 ● < 0.05
Convergent EC
Pathway
Species